Lab Report: 4

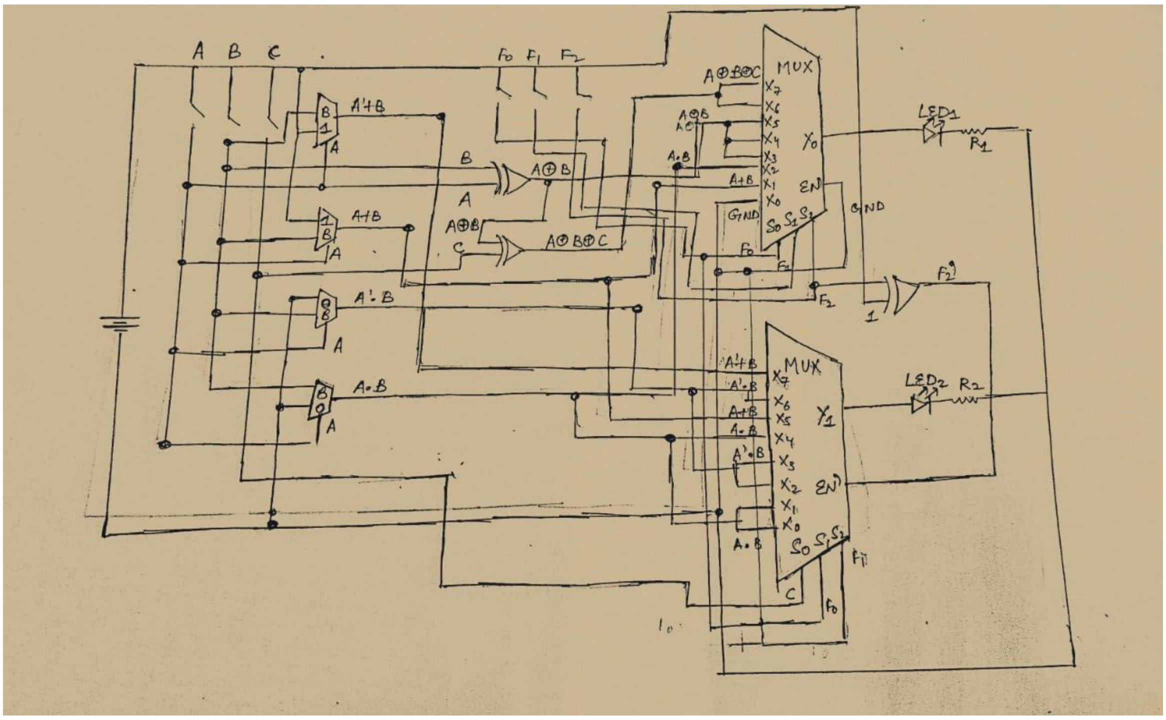
Name: Saarthak Sabharwal
Roll number: 2023102055
Group number: 6

Objective: Designing, assembling and testing a programmable ALU

Electronic components used:

- Digital test kit (w/ breadboard)
- 2x 74LS157 ICs
- 2x 74LS151 ICs
- 1x 74LS86 IC
- Arduino Uno
- Wires

The reference circuit:

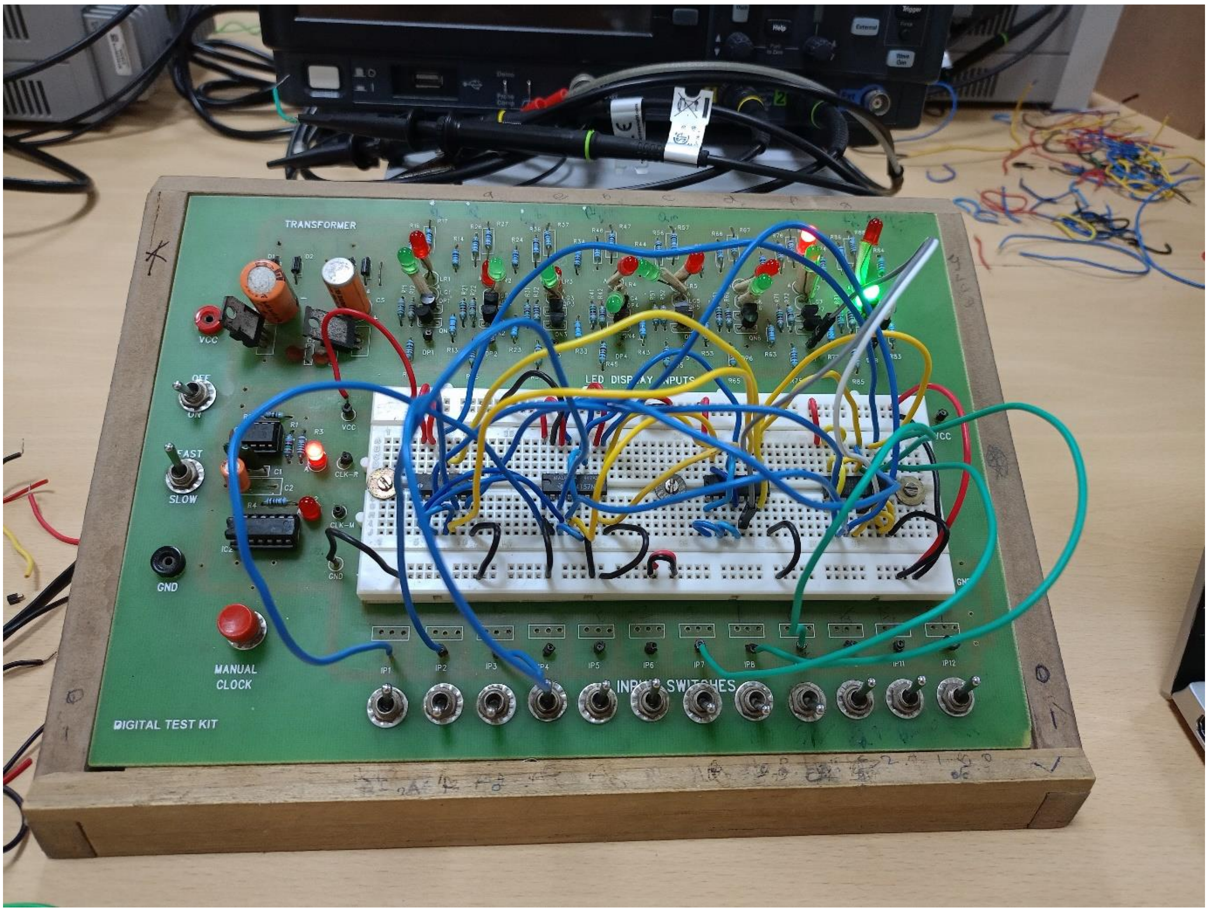


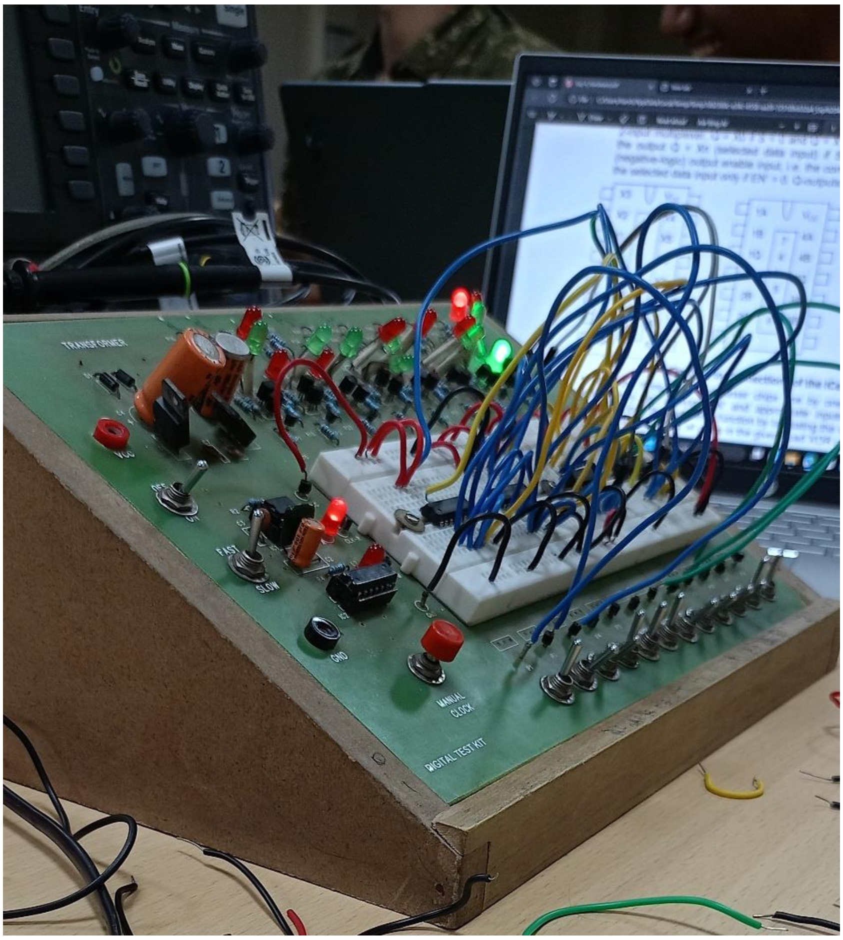
Arduino Code:

```
//Auth: Saarthak_2023102055
//ALU controller
//input format:
//  F2|F1|F0|A|B|C 0|1

void setup()
{
  for(int i = 8; i <= 13; i++)
    pinMode(i, OUTPUT);
  Serial.begin(9600);
}

void loop()
{
  while(!Serial.available());

  String cmd = Serial.readString();
  Serial.println(cmd);

  switch(cmd[0]) {
    case 'F':
        digitalWrite(cmd[1]-37, cmd[3]);
        break;
    default:
        digitalWrite(cmd[0]-57, cmd[2]);
  }
}
```

Procedure:

- Plug each IC into the breadboard and test it for correct outputs.
- Now, set up the circuit on the breadboard as shown in diagram.
- Power the circuit through the Arduino and provide inputs through digital pins 8-13, as shown, instead of switches.
- Via the serial monitor, provide all possible input combinations of function select inputs (F_0, F_1, F_2) and tabulate the outputs for each function into a truth table.


TRANSFORMER
VCC
OFF
ON
FAST
SLOW
GND
MANUAL
CLOCK
DIGITAL TEST KIT
INPUT SWITCHES


TRANSFORMER
GND
MANUAL
CLOCK
DIGITAL TEST KIT

Conclusion: Following are the truth tables for each of the 8 functions within the Arithmetic and Logic Unit (ALU):

<table>
<tr><th>$F_2 F_1 F_0$</th><th>Truth Table</th></tr>
<tr><td>0 0 0</td><td><table><tr><th>A</th><th>B</th><th>Y_0</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table></td></tr>
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</table>

1 1 0

A	B	C	Y_1	Y_0
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

1 1 1

A	B	C	Y_1	Y_0
0	0	0	0	0
0	0	1	1	1
0	1	0	0	1
0	1	1	0	0
1	0	0	0	1
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

Link for the Tinkercad simulation:

https://www.tinkercad.com/things/aFbHW1gXoCB-lab4/editel?sharecode=NeUcc32gJVOGpSY7o_Ctqw_505BL33xGvWPSbapFv50